Comparative Analysis of Gene Expression in Surgically Removed Lung Adenocarcinoma and Normal Adjacent Lung Tissues.

Bobby Luker

April 28, 2026

Introduction:

Lung adenocarcinoma is a type of non-small cell lung cancer (NSCLC), which typically forms within the alveoli where gas exchange occurs. Lung adenocarcinoma accounts for nearly 40% of all NSCLC and is the most frequent lung cancer in non-smokers (“Adenocarcinoma of the Lung”). Improving our understanding of the genetic mutations and mechanisms in lung adenocarcinoma is essential for advancing diagnostics and therapeutics for NSCLC.

Utilizing RNA sequencing (RNA-seq) is a key tool for understanding significant transcriptomic changes between tumor and normal tissues. Previous differential gene expression and pathway-based analyses using RNA-seq data have identified key biological processes involved in lung adenocarcinoma, including epithelial-to-mesenchymal transition (EMT), extracellular matrix remodeling, and dysregulation of cell cycle pathways (Kalluri & Weinberg, 2009). These approaches provide insight into both tumor biology and potential molecular targets for therapeutic development.

Environmental exposures, particularly tobacco smoking, also heavily influence the lung cancer transcriptome. It has been previously reported that tobacco-related NSCLC exhibits a 10-fold increase in mutational frequency in smokers compared to non-smokers (Govindan et al., 2012). In this study, RNA-seq data from the GSE40419 dataset were analyzed to characterize differential gene expression between lung adenocarcinoma tissue and matched normal lung tissue. Additionally, the impact of smoking status on tumor-associated gene expression was evaluated through interaction modeling and subgroup analyses to better understand its role in lung adenocarcinoma mechanisms.

Materials and Methods:

Using the GSE40419 dataset, RPKM-normalized RNA-seq data for 87 lung adenocarcinomas and 77 normal lung tissue samples from a Korean population were analyzed for differential gene expression. Table 1 summarizes the sample sizes by smoker status and tissue type for all 164 samples. The 87 lung adenocarcinoma samples “were analyzed by transcriptome sequencing combined with whole-exome (n = 76) and transcriptome (n = 77) sequencing of matched normal lung tissue samples” (Seo et al., 2012).

In this study, the RNA-seq data were cleaned and filtered for downstream analysis. Specifically, duplicate gene symbols were aggregated by taking the mean expression across entries. Genes were then filtered to retain those with expression values greater than one in at least 20% of samples, removing lowly expressed genes. Then, the data were log2-transformed with a pseudo-count of one.

Following the filtration and transformation, principal component analysis (PCA) was performed on the scaled log2-transformed expression values of tissue groups by smoking status. Differential gene expression analysis was then performed on paired samples, including an analysis of the interaction between smoking status and tissue type. Only patients with both tumor and matched normal samples were retained (n = 77 pairs). For paired analyses, patient identity was included as a blocking factor in the design matrix (~ Patient + Tissue). For interaction analysis, the linear model included a tissue-by-smoking interaction term by patient identity (~ Patient + Tissue × Smoking_Group) to evaluate whether smoking status impacts tumor-associated gene expression changes. For subgroup analyses, contrast-based linear models were used to estimate the tissue type differences within each smoking status group. Additionally, the subgroup analyses were not restricted to the paired sample size and instead used the appropriate subgroup sample size as seen in Table 1.

All differential gene expression analyses were performed using the *limma* package in R. All linear models were fitted to the log2-transformed expression data, and statistical significance was assessed using empirical Bayes moderated t-tests to stabilize variance estimates. P-values for all models were adjusted for multiple testing using the Benjamini–Hochberg (BH) false discovery rate (FDR) method. The top ten upregulated and downregulated differentially expressed genes (DEGs) were also identified for each model. The top three upregulated and downregulated DEGs for each model were incorporated into Table 2. P-value histograms and QQ plots were used to assess statistical calibration for all models as well.

After the paired analysis, gene symbols were mapped to Entrez IDs using org.Hs.eg.db, and Gene Ontology (GO) enrichment analysis was performed using a hypergeometric test, with all tested genes as the background universe. Following the smoking interaction analysis, Gene Set Enrichment Analysis (GSEA) was performed with genes ranked by interaction log2 fold change to identify biological pathways associated with tumor-specific transcriptional changes.

OpenAI's ChatGPT model GPT-5.3-mini was used to guide the creation of R code to generate figures, perform PCA and GSEA analysis, and use of *limma* for differential gene expression analysis. The code for this project can be found at this link: https://github.com/bluker17/Lung_Adenocarcinoma_Analysis

Results:

Global Transcriptomic Profile

Principal component analysis (PCA) revealed clear separation between lung adenocarcinoma and normal lung tissue samples (Figure 1). The first two principal components together accounted for 30.4% of the total variance (PC1 = 20.1%, PC2 = 10.3%). Tumor samples clustered distinctly from normal samples, with partial overlap observed across smoking and non-smoking subgroups.

Paired Patient Sample Differential Gene Expression Analysis

In the primary patient tissue sample paired analysis (n = 77 matched pairs), 1,265 genes were significantly differentially expressed between tumor and normal lung tissue (FDR < 0.05, |log2FC| > 1). Of these, 467 were upregulated and 798 were downregulated in tumor tissue (Figure 2). The three most upregulated DEGs were *SPP1*, *PYCR1*, and *ALDH18A1*. The three most downregulated DEGs were *GPM6A*, *CCBE1*, and *RADIL* (Table 2).

Subsequent gene ontology enrichment analysis of significant DEGs identified multiple significantly enriched biological processes related to angiogenesis regulation, cellular migration, and extracellular matrix organization (Figure 3).

Smoking-Dependent Interaction Effects

Interaction analysis between smoking status and tissue type identified 28 genes with significant smoking-dependent effects on tumor-associated gene expression (FDR < 0.05). Nine genes showed positive interaction effects associated with smoking, while 19 showed negative interaction effects (Figure 4). The most significant upregulated genes included *CDC20*, *PRAME*, and *MYBL2*, while the most significant downregulated genes included *SCTR*, *C1orf116*, and *ABCC3* (Table 2).

GSEA revealed that genes involved in biological processes, such as chromatid and chromosome segregation, and nuclear division, were positively enriched in tumor tissue compared to normal tissue (Figure 5).

Exploratory Differential Gene Analysis

In all patient samples (n_{tumor} = 87, n_{normal} = 77), 1,300 DEGs between tumor and normal lung tissue were identified (FDR < 0.05, |log2FC| > 1). Of these, 487 were upregulated and 813 were downregulated in tumor tissue (Figure 6).

In non-smokers ($n_{tumor\_non\text{-}smoker}$=36, $n_{normal\_non\text{-}smoker}$=34), 1,136 genes were differentially expressed between non-smoker tumor and normal lung tissue (FDR < 0.05, |log2FC| > 1). Of these, 438 genes were upregulated and 698 genes were downregulated in tumor tissue (Figure 7).

In smokers (n_{tumor_smoker}=47, n_{normal_smoker}=40), 1,559 genes were differentially expressed between smoker tumor and normal lung tissue (FDR < 0.05, |log2FC| > 1). Of these, 563 genes were upregulated and 996 genes were downregulated in tumor tissue (Figure 8).

The smoker subgroup showed a greater number of DEGs compared to the non-smoker subgroup. This may reflect an increased gene mutational frequency in tumors associated with smokers compared to non-smokers.

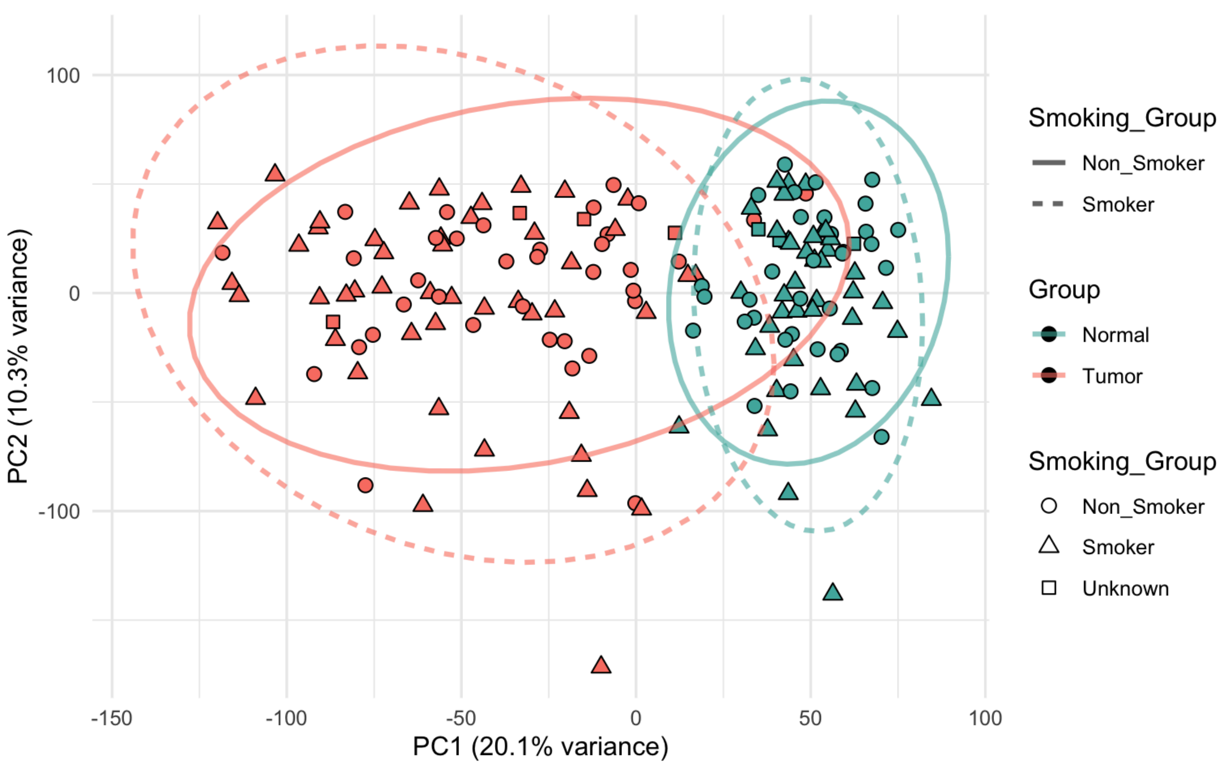

Figure 1. PCA reveals distinct transcriptome patterns of lung adenocarcinoma tissue and normal adjacent lung tissue. PCA plot of RNA-seq data from lung adenocarcinoma tissue and

normal adjacent lung tissue (n_{tumor_smoker}=47, $n_{tumor\_non\text{-}smoker}$=36, $n_{tumor\_unknown}$=4, n_{normal_smoker}=40, $n_{normal\_non\text{-}smoker}$=34, $n_{normal\_unknown}$=3). Each point represents an individual sample, teal points indicate the normal sample population, and red points indicate the tumor sample population. Triangles indicate the smoker sample population, circles the non-smoker sample population, and squares the unknown sample population. PC1 accounts for 20.1% of the variance between groups. PC2 accounts for 10.3% of the variance within groups. Dashed ellipses indicate the approximate distribution and spread of each group.

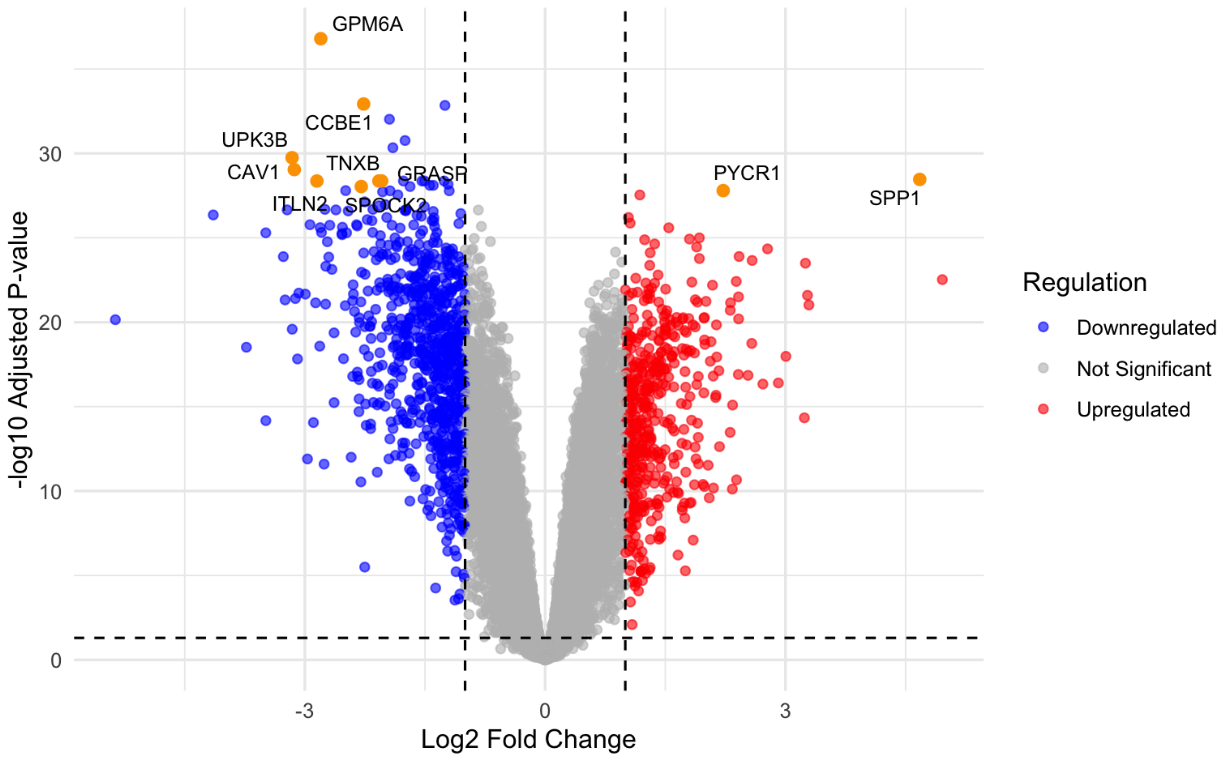

Figure 2. Volcano plot reveals DEGs of paired patient lung adenocarcinoma tissue and normal adjacent lung tissue samples. Volcano plot of log2-transformed RNA-seq data from paired lung adenocarcinoma tissue and normal adjacent lung tissue (n_{pairs}=77). Paired differential expression was performed using *limma* with a linear model (~ patient + group) and empirical Bayes moderated t-tests, with BH FDR correction. Each point represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression ($|log2FC| > 1$ and adjusted p-value < 0.05).

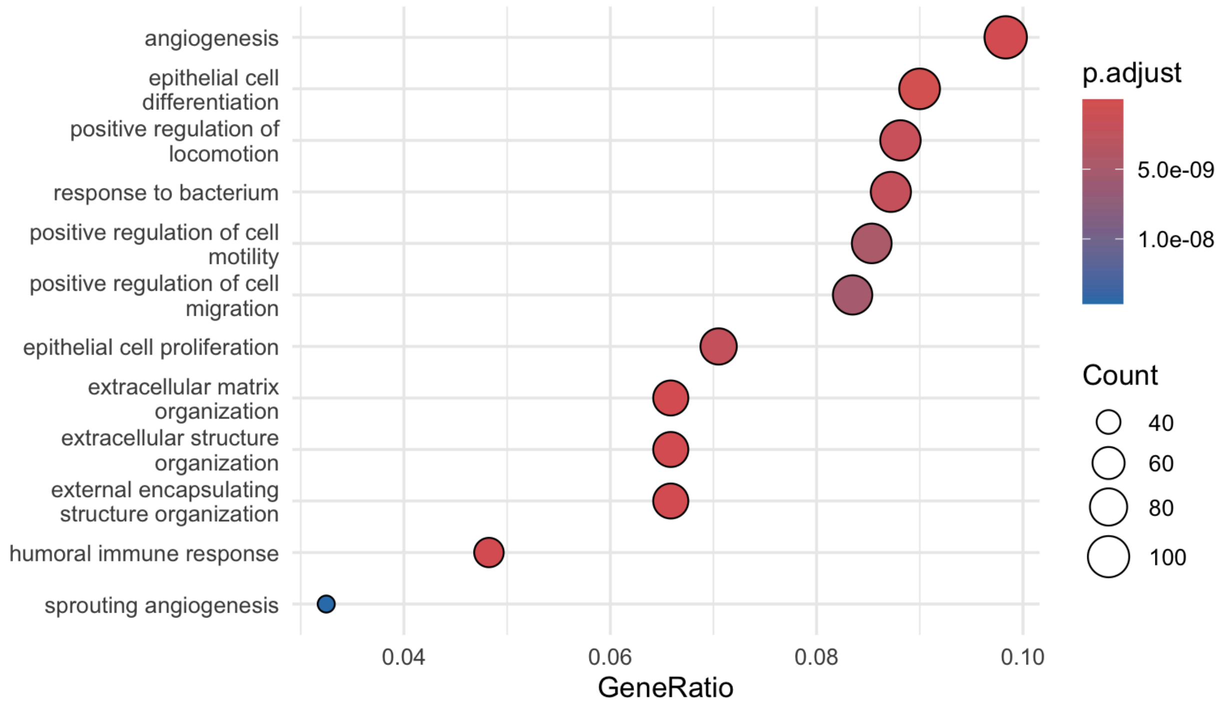

Figure 3. Gene ontology enrichment analysis of biological processes associated with paired tumor and normal tissue DEGs. GO enrichment analysis was performed on significantly DEGs identified from the paired lung adenocarcinoma vs normal adjacent lung tissue comparison. Paired differential expression was performed using *limma* with a linear model (~ patient + group) and empirical Bayes moderated t-tests, with BH FDR correction. Enrichment of Gene Ontology Biological Process (BP) terms was tested using a hypergeometric overrepresentation test, with all genes tested in the paired model used as the background universe. Each term represents a GO Biological Process category. The top 12 enriched terms are displayed. The color scale represents adjusted p-value significance (BH-FDR), with more significant terms shown in warmer colors.

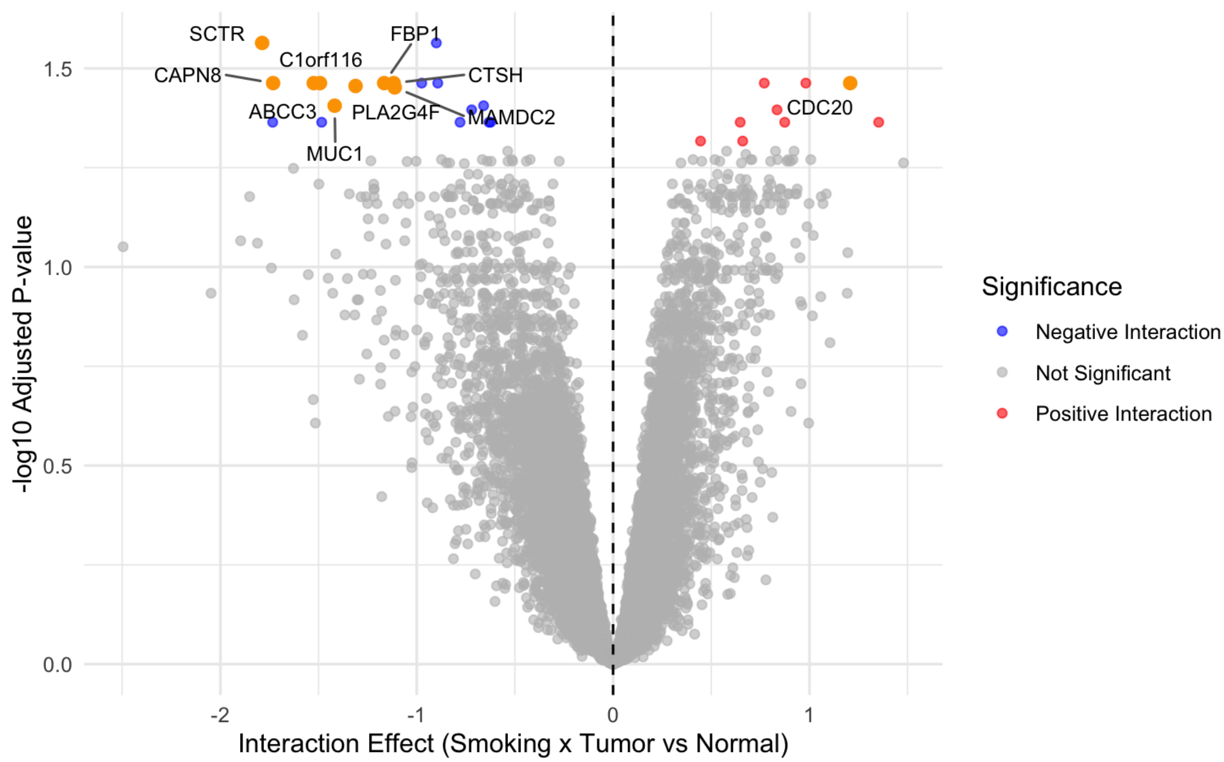

Figure 4. Volcano plot reveals genes with smoking-dependent tumor-associated expression changes in paired lung adenocarcinoma and normal adjacent lung tissue samples. Volcano plot of log2-transformed RNA-seq data from paired lung adenocarcinoma tissue and normal adjacent lung tissue (n_{pairs}=77). Interaction analysis was performed using *limma* with a linear model (~ Patient + Tissue × Smoking_Group) and empirical Bayes moderated t-tests, with BH FDR correction. Each dot represents an individual gene. Red dots represent genes with positive interaction effects (greater tumor-associated expression changes in smokers), blue dots represent genes with negative interaction effects (reduced or reversed tumor-associated expression changes in smokers), and gray dots represent non-significant genes. Orange dots represent the ten most significant interaction genes. Dashed line marks the cutoff for no interaction effect ($|log2FC| > 0$ and adjusted p-value < 0.05).

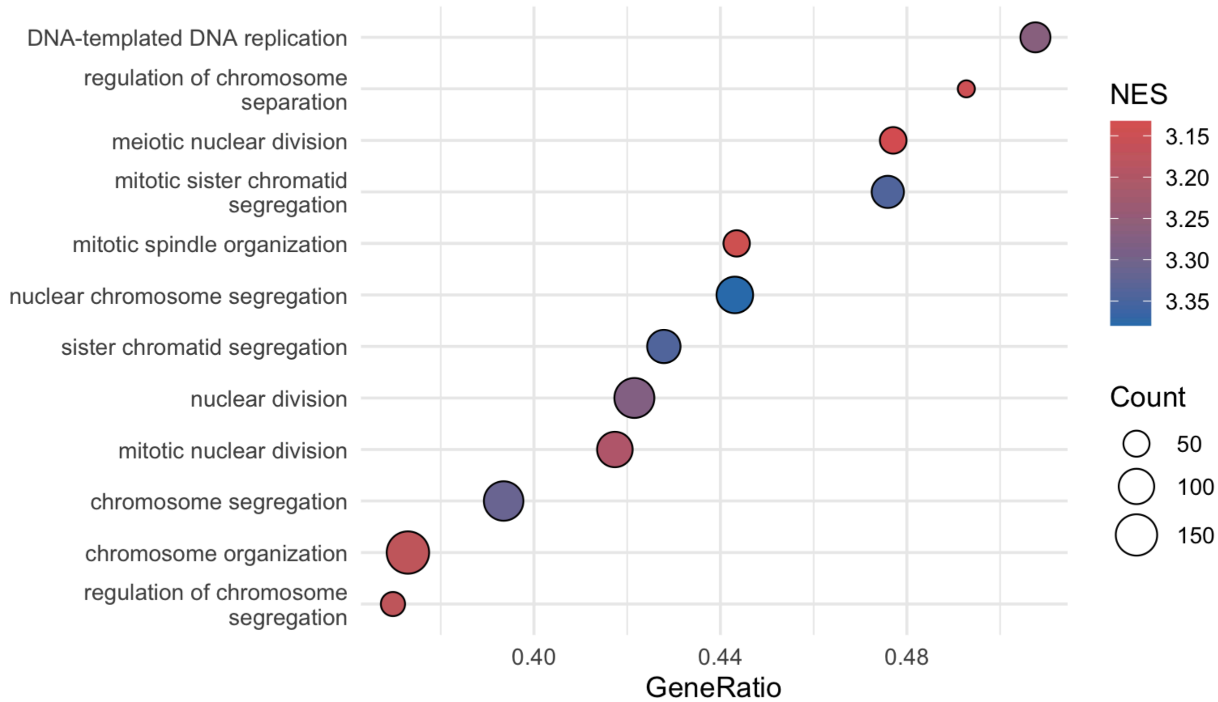

Figure 5. Gene set enrichment analysis of biological processes associated with the smoking–tissue interaction. Gene set enrichment analysis (GSEA) was performed on a ranked list of genes ordered by the interaction log2 fold change derived from a paired differential expression analysis of lung adenocarcinoma and matched normal adjacent tissue. Differential expression was modeled using *limma* with a linear model (~ patient + group × smoking status) and empirical Bayes moderated t-tests, with p-values adjusted using the BH FDR. Enrichment of Gene Ontology Biological Process (BP) terms was assessed using a permutation-based GSEA approach implemented in *clusterProfiler*. The top 12 enriched terms are shown. Color represents the normalized enrichment score (NES), with cooler color indicating more positive enrichment.

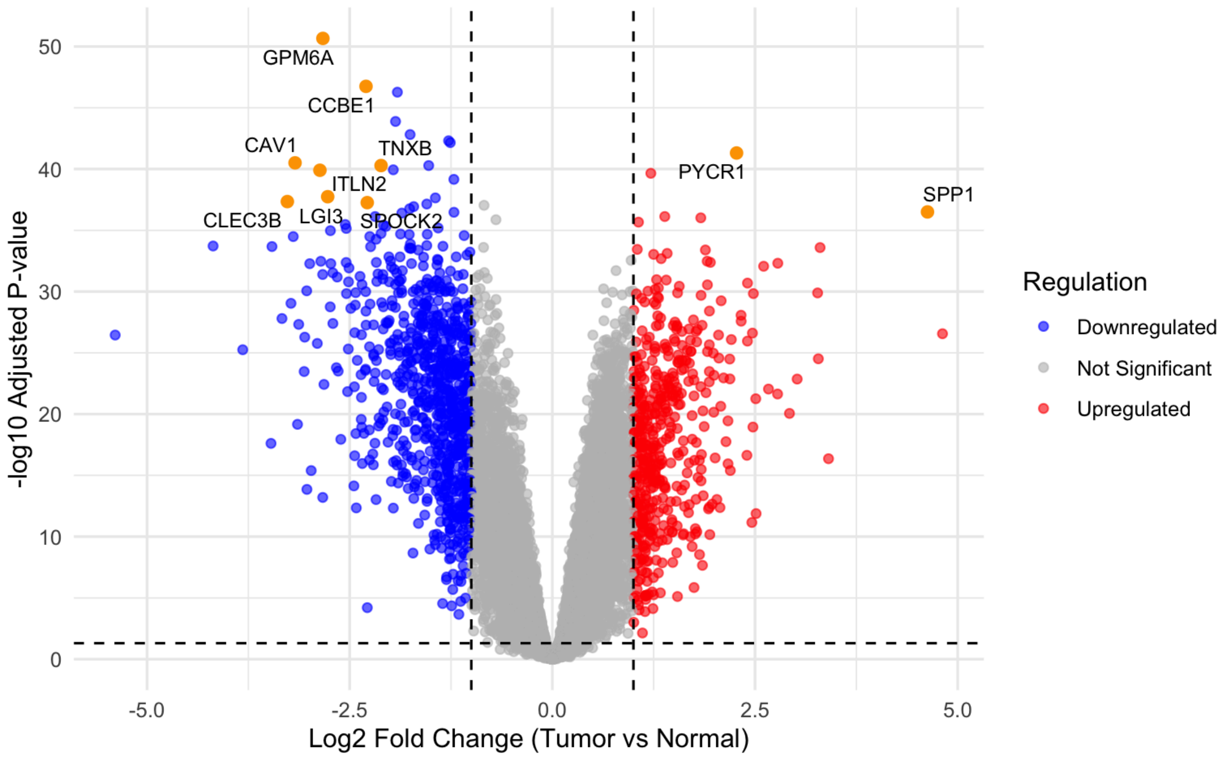

Figure 6. Volcano plot reveals DEGs of lung adenocarcinoma tissue and normal adjacent lung tissue. Volcano plot of log2-transformed RNA-seq data from lung adenocarcinoma tissue and normal adjacent lung tissue (n_{tumor}=87, n_{normal}=77). Differential expression analysis was performed using *limma* with a linear model (~ group) and empirical Bayes moderated t-tests, with BH FDR correction. Each dot represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression (|log2FC| > 1 and adjusted p-value < 0.05).

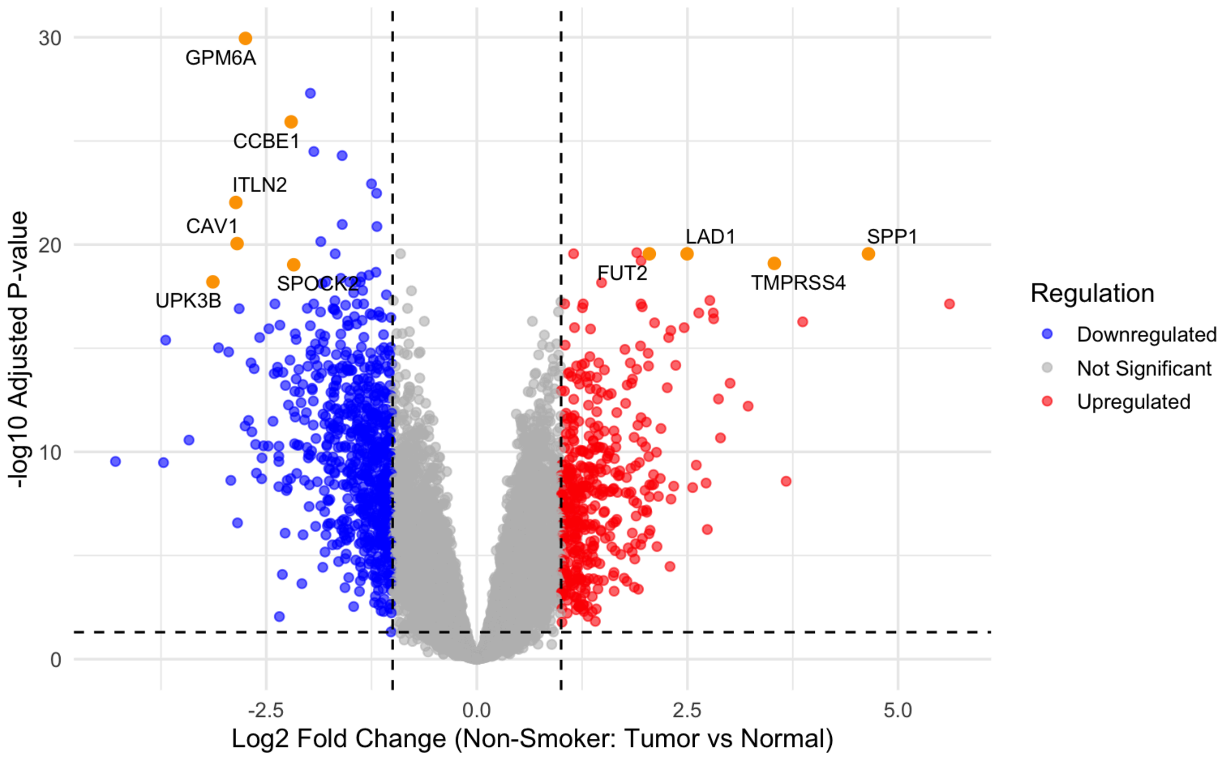

Figure 7. Volcano plot reveals DEGs of non-smoker lung adenocarcinoma tissue and non-smoker normal adjacent lung tissue. Volcano plot of log2-transformed RNA-seq data from non-smoker lung adenocarcinoma tissue and non-smoker normal adjacent lung tissue ($n_{tumor\_non-smoker}$=36, $n_{normal\_non-smoker}$=34). Differential expression analysis was performed using *limma* with a contrast-based linear model (~ 0 + Group_Non_Smoking) and empirical Bayes moderated t-tests applied to the Tumor vs Normal contrast within non-smoker samples, with BH FDR correction. Each point represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression ($|log2FC| > 1$ and adjusted p-value < 0.05).

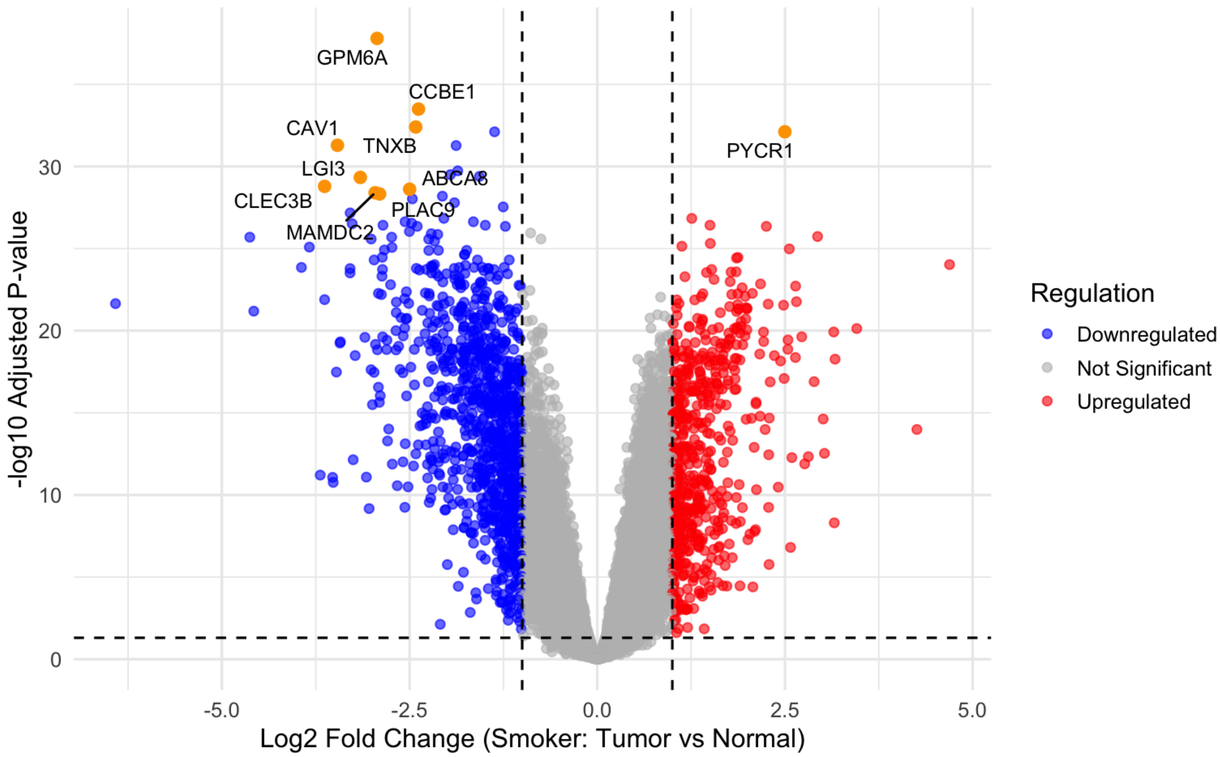

Figure 8. Volcano plot reveals DEGs of smoker lung adenocarcinoma tissue and smoker normal adjacent lung tissue. Volcano plot of log2-transformed RNA-seq data from smoker lung adenocarcinoma tissue and smoker normal adjacent lung tissue (n_{tumor_smoker}=47, n_{normal_smoker}=40). Differential expression analysis was performed using *limma* with a contrast-based linear model (~ 0 + Group_Smoking) and empirical Bayes moderated t-tests applied to the Tumor vs Normal contrast within smoker samples, with BH FDR correction. Each point represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression (|log2FC| > 1 and adjusted p-value < 0.05).

Table 1: Sample Sizes by Group and Smoking Status

Group	Non_Smoker	Smoker	Unknown
Normal	34	40	3
Tumor	36	47	4

Table 2: Top upregulated and downregulated genes identified in tumor vs normal analyses across all samples, paired samples, smokers, non-smokers, and smoking interaction analysis (* indicates genes significant in multiple comparisons)

Comparison	Regulation	gene	logFC
Tumor vs Normal (All)	Upregulated	PYCR1*	2.272296
Tumor vs Normal (All)	Upregulated	ALDH18A1*	1.214749
Tumor vs Normal (All)	Upregulated	SPP1*	4.627416
Tumor vs Normal (All)	Downregulated	GPM6A*	-2.830690
Tumor vs Normal (All)	Downregulated	CCBE1*	-2.298876
Tumor vs Normal (All)	Downregulated	C13orf36*	-1.911831
Tumor vs Normal (Non-smokers)	Upregulated	GOLM1	1.897995
Tumor vs Normal (Non-smokers)	Upregulated	ALDH18A1*	1.148396
Tumor vs Normal (Non-smokers)	Upregulated	FUT2	2.047086
Tumor vs Normal (Non-smokers)	Downregulated	GPM6A*	-2.746787
Tumor vs Normal (Non-smokers)	Downregulated	C13orf36*	-1.976093
Tumor vs Normal (Non-smokers)	Downregulated	CCBE1*	-2.204655
Tumor vs Normal (Smokers)	Upregulated	PYCR1*	2.498757
Tumor vs Normal (Smokers)	Upregulated	ALDH18A1*	1.259134
Tumor vs Normal (Smokers)	Upregulated	RCC1	1.501677
Tumor vs Normal (Smokers)	Downregulated	GPM6A*	-2.933016
Tumor vs Normal (Smokers)	Downregulated	CCBE1*	-2.380980
Tumor vs Normal (Smokers)	Downregulated	TNXB	-2.418220
Paired	Upregulated	SPP1*	4.675954
Paired	Upregulated	PYCR1*	2.222348
Paired	Upregulated	ALDH18A1*	1.181269
Paired	Downregulated	GPM6A*	-2.801642
Paired	Downregulated	CCBE1*	-2.266637
Paired	Downregulated	RADIL	-1.251265
Paired Smoker Interaction	Upregulated	CDC20	1.207580
Paired Smoker Interaction	Upregulated	MYBL2	1.352700
Paired Smoker Interaction	Downregulated	SCTR	-1.787325
Paired Smoker Interaction	Downregulated	C1orf116	-1.492882
Paired Smoker Interaction	Downregulated	ABCC3	-1.524474

Discussion:

This study identified widespread differences in gene expression between lung adenocarcinoma and normal adjacent lung tissue using RPKM-normalized RNA-seq data from the GSE40419 dataset. Principal component analysis demonstrated clear separation between tumor and normal samples, indicating distinct global transcriptional profiles between tumor and normal lung tissue.

Differential gene expression analysis in paired samples identified 1,265 DEGs, with a greater proportion of genes downregulated in tumor tissue, which suggests a loss of normal lung function. Using a paired design allowed for direct comparison of tumor and normal samples from the same patient, reducing variation between individuals. GO enrichment analysis identified biological processes related to angiogenesis regulation, cell migration, and extracellular matrix organization. These pathways were reflected in highly upregulated genes such as *SPP1* and *PYCR1*. *SPP1* has been widely reported across multiple cancers, including lung cancer, and has been associated with increased cell proliferation and angiogenesis through promotion of EMT (Tang et al., 2021). EMT is a biological process in which epithelial cells acquire mesenchymal-like properties, including increased migratory capacity, extracellular matrix production, and resistance to apoptosis (Kalluri & Weinberg, 2009). *PYCR1* has also been implicated in tumor progression and may contribute to increased cell proliferation and survival through activation of the JAK/STAT signaling pathway, which regulates cellular differentiation, proliferation, and apoptosis (Shin et al., 2025).

Another key finding of this study was the identification of 28 genes with significant smoking-dependent effects identified in the interaction analysis. This limited number suggests that smoking may not globally reshape tumor-associated transcriptomes, but it may affect specific tumor-associated genetic mechanistic pathways. However, the limited number of DEGs suggests that larger sample sizes are needed to improve sensitivity for detecting smoking-dependent genetic effects. Although few DEGs were identified, several are involved in cell cycle regulation and proliferation, which is consistent with GSEA. For example, *CDC20*, one of the most significant upregulated genes identified in the interaction analysis, has been implicated in cell cycle regulation and may contribute to tumor progression through effects on proliferation and apoptotic regulation in lung cancer (Xian et al., 2023).

Smoking subgroup analyses further demonstrated that smokers exhibited a higher number of differentially expressed genes compared to non-smokers. This observation is consistent with the established association between tobacco exposure and increased genetic mutations in lung cancer (Govindan et al., 2012). This suggests that smoking may contribute to these increased mutations or enhanced molecular divergence between tumor and normal lung tissue. However, it is important to note that these subgroup analyses were not fully paired and are more susceptible to confounding factors.

A few limitations of this study should be considered when interpreting these results. First, the dataset is derived from a single population, which may limit generalizability to other ethnic or demographic groups. Second, smoking status was categorized broadly and did not account for exposure intensity or duration, which may make it more difficult to detect specific effects of tobacco use on gene expression. Third, the use of RPKM-normalized RNA-seq data, rather than raw count-based normalization methods, may introduce biases in expression estimates and affect differential expression results.

For future lung adenocarcinoma analysis, collecting patient data where smoking is categorized by time or smoking intensity would improve the conclusions on the impact of smoking on tumor development. Additionally, incorporating a machine learning model, such as random forest, may help reveal predictive carcinogenic genes to extend beyond DEG analysis. Depending on the machine learning model's accuracy, it could reveal potential genes of interest to monitor for diagnostic testing.

References:

"Adenocarcinoma of the Lung." *Cancer Treatment Centers of America*, reviewed by Peter Baik, DO, Cancer Treatment Centers of America, 7 Mar. 2023, https://www.cancercenter.com/cancer-types/lung-cancer/types/adenocarcinoma-of-the-lung

GEO Accession Dataset - GSE40419

Govindan, R., Ding, L., Griffith, M., Subramanian, J., Dees, N. D., Kanchi, K. L., ... & Wilson, R. K. (2012). Genomic landscape of non-small cell lung cancer in smokers and never-smokers. *Cell, 150*(6), 1121–1134. https://doi.org/10.1016/j.cell.2012.08.024

Kalluri R, Weinberg RA. The basics of epithelial-mesenchymal transition. *J Clin Invest*. 2009 Jun;119(6):1420-8. doi: 10.1172/JCI39104. Erratum in: J Clin Invest. 2010 May 3;120(5):1786. PMID: 19487818; PMCID: PMC2689101.

OpenAI. *ChatGPT, GPT-5.3-mini*. OpenAI, March 2026, https://chat.openai.com/

Seo JS, Ju YS, Lee WC, Shin JY et al. The transcriptional landscape and mutational profile of lung adenocarcinoma. *Genome Res* 2012 Nov;22(11):2109-19. PMID: 22975805

Shin, J.H., Kim, J.Y., Kim, MJ. et al. PYCR1 drives lung cancer progression through functional interactions with EGFR and TLR signaling pathways. *Exp Mol Med* **57**, 2559–2573 (2025). https://doi.org/10.1038/s12276-025-01577-z

Tang H, Chen J, Han X, Feng Y, Wang F. Upregulation of *SPP1* Is a Marker for Poor Lung Cancer Prognosis and Contributes to Cancer Progression and Cisplatin Resistance. *Front Cell Dev Biol*. 2021 Apr 29;9:646390. doi: 10.3389/fcell.2021.646390. PMID: 33996808; PMCID: PMC8116663.

Xian F, Zhao C, Huang C, Bie J, Xu G. The potential role of CDC20 in tumorigenesis, cancer progression and therapy: A narrative review. *Medicine (Baltimore)*. 2023 Sep 8;102(36):e35038. doi: 10.1097/MD.0000000000035038. PMID: 37682144; PMCID: PMC10489547.